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B.Tech. Degree VII Semester Regular/Supplementary Examination November 2024

ME 19-205-0706 AUTOMOBILE ENGINEERING (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Compare the working of various types of automobiles.
 CO2: Learn the functions and working of different engine components.
 CO3: Appreciate the use of alternate fuels.
 CO4: Understand the latest technology used in the field of automobile.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	Differentiate between wet cylinder liner and dry cylinder liner.	3	L2	2	1,2
(b)	What do you mean by valve overlap? Discuss its effect on low speed and high speed engines.	3	L2	4	1,2
(c)	What are the different types of air cooling systems available for automobiles?	3	L2	2	1,2
(d)	What are the advantages of using CNG as an alternate fuel?	3	L2	3	1,2
(e)	What are the advantages of using CVT transmission?	3	L2	2	1,2
(f)	Explain the working of an overdrive with the help of a neat sketch.	3	L2	2	1,2
(g)	Explain the configuration and control of DC motor drives.	3	L2	4	1,2
(h)	Explain the principle of flywheel based energy storage in an automobile.	3	L2	3	1,2

PART B

(4 × 12 = 48)

II.	(i) What is the composition of Lo-Ex alloy? State its application in automobiles.	12	L2	2	1,2
	(ii) What is the application of spheroidal graphite cast iron in engine construction?				
	(iii) Explain the significance of wheel balancing in automobiles.				
	(iv) Explain the term valve clatter. When does it occur?				

OR

III.	(a) Compare different valve actuating mechanisms in automobile engines with neat sketches.	8	L2	4	1,2
	(b) What are the exhaust valve material requirements?	4			

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		Marks	BL	CO	PO
IV.	What do you mean by compensation as applied to carburettors? Discuss various methods of compensation with neat sketches.	12	L2	2	1,2
OR					
V. (a)	Why a starting circuit is a necessity for the proper working of an IC engine?	2	L3	2	1,2
(b)	Discuss various arrangements of starting circuits in automobiles with neat sketches.	10	L2	2	1,2
VI. (a)	What do you mean by double declutching? State its advantages.	2	L2	1	1,2
(b)	Discuss the working of synchromesh gear box with neat sketches.	10	L2	2	1,2
OR					
VII. (a)	What do you mean by wheel alignment? What are the factors affecting wheel alignment?	2	L2	4	1,2
(b)	Discuss the steering geometry with neat sketches. Also discuss its effect on stability.	10	L2	2	1,2
VIII.	Why hybrid engines are getting wider acceptability over IC engines? Compare the advantages and disadvantages.	12	L2	4	1,2
OR					
IX.	Compare the working of an electric car with an IC engine with the help of a block diagram.	12	L2	4	1,2

Blooms's Taxonomy Levels

L2 – 90%, L3 – 10%.

